

NCC VOLUME TWO, ENERGY EFFICIENCY

Group Exercise Handout

EXERCISE 1 – Minimum Total R-Value

QUESTION 1

In your own words, describe the meaning of R-Value or material R-Value?

QUESTION 2

In your own words, describe the difference between R-Value and Total R-Value?

QUESTION 3

For what elements of a house, and in which clauses, do the NCC energy efficiency Deemed-to-Satisfy Provisions prescribe minimum Total R-Values?

QUESTION 4

What is the minimum Total R-Value that a roof and ceiling in climate zone 1 is required to achieve?

QUESTION 5

What is the minimum Total R-Value required for a roof with a flat ceiling in climate zone 2 (below 300 m altitude) and a roof solar absorptance of 0.55?

QUESTION 6

What is the inherent Total R-Value of an un-insulated pitched metal roof with a flat ceiling in climate zone 2 (below 300 m altitude) and in climate zone 6?

QUESTION 7

How much insulation is required to be added to a pitched dark tiled roof with a flat ceiling in climate zone 6?

QUESTION 8

How much insulation is required to be added to a pitched metal roof with a flat ceiling and ventilated roof space in climate zone 2 (below 300 m altitude)?

QUESTION 9

What needs to be considered when selecting insulation?

QUESTION 10

What needs to be considered when installing insulation?

QUESTION 11

What also needs to be considered when installing reflective insulation?

Worked Example

To answer the following questions you will need to know the following additional information.

- The building is in climate zone 3.
- The floor area is 100 m².
- The direction of heat flow is downward and upward.
- The roof is red and has a solar absorptance value of 0.75.
- The roof cladding is metal.
- The roof space is unventilated.
- There is one service opening in the insulation with an area of 1 m².
- There are 11 opening in the insulation for lights.
- The lights are 100 mm wide in diameter.

QUESTION 12

What is the uninsulated R-Value of the roof and the minimum added insulation required?

QUESTION 13

What is the percentage of insulation lost from the service opening and light installations?

QUESTION 14

What is the adjusted minimum R-Value of ceiling insulation required to compensate for the loss of ceiling insulation?

EXERCISE 2 – Glazing

QUESTION 1

In your own words, what constitutes the external glazing of a house?

QUESTION 2

In your own words, describe the meaning of Total System U-Value?

QUESTION 3

In your own words, describe the meaning of Total System Solar Heat Gain Coefficient (SHGC)?

QUESTION 4

Using 3.12.2.1(a), determine whether the external glazing for the house described below, in conjunction with the floor plan and glazing schedule on the next page, is compliant.

- House is located in climate zone 5.
- Single storey slab-on-ground concrete floor.
- 100 m² rectangular floor plan, long axis east - west.
- 300 mm eaves on all sides.
- Window head height 100 mm below the eaves.
- Standard ventilation.
- Single glazed clear aluminum windows
(Total System U-Value of 7.9 W/m².K and Total System SHGC of 0.81).

GLAZING SCHEDULE

GLAZING SCHEDULE	
G1	1800h x 2400w Sliding Window
G2	1200h x 2400w Sliding Window
G3	1800h x 2400w Sliding Window
G4	1800h x 2400w Sliding Window
G5	2100h x 2400w Sliding Door
G6	1800h x 2400w Sliding Window
G7	1800h x 2400w Sliding Window
G8	1800h x 2400w Sliding Window

HOUSE PLAN

